Evidence for acne-promoting effects of milk and other insulinotropic dairy products.

Acne vulgaris, the most common skin disease of western civilization, has evolved to an epidemic affecting more than 85% of adolescents. Acne can be regarded as an indicator disease of exaggerated insulinotropic western nutrition. Especially milk and whey protein-based products contribute to elevations of postprandial insulin and basal insulin-like growth factor-I (IGF-I) plasma levels. It is the evolutional principle of mammalian milk to promote growth and support anabolic conditions for the neonate during the nursing period. Whey proteins are most potent inducers of glucose-dependent insulinotropic polypeptide secreted by enteroendocrine K cells which in concert with hydrolyzed whey protein-derived essential amino acids stimulate insulin secretion of pancreatic β-cells. Increased insulin/IGF-I signaling activates the phosphoinositide-3 kinase/Akt pathway, thereby reducing the nuclear content of the transcription factor FoxO1, the key nutrigenomic regulator of acne target genes. Nuclear FoxO1 deficiency has been linked to all major factors of acne pathogenesis, i.e. androgen receptor transactivation, comedogenesis, increased sebaceous lipogenesis, and follicular inflammation. The elimination of the whey protein-based insulinotropic mechanisms of milk will be the most important future challenge for nutrition research. Both, restriction of milk consumption or generation of less insulinotropic milk will have an enormous impact on the prevention of epidemic western diseases like obesity, diabetes mellitus, cancer, neurodegenerative diseases and acne.